ZZ-T-381Q
August 1, 1989
SUPERSEDING
ZZ-T-381P
August 1, 1983

FEDERAL SPECIFICATION

TIRES, PNEUMATIC, VEHICULAR (HIGHWAY)

This Federal Specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 SCOPE. This specification covers pneumatic tires, new and retreaded, both tube type and tubeless, for use on conventional passenger cars, station wagons, pursuit and emergency high speed and pursuit passenger vehicles, trucks, trailers, buses, and similar vehicles normally operated on public roads and highways.

1.2 CLASSIFICATION.

1.2.1 STYLES, GROUPS, TYPES AND CLASSES. Tires shall be of the following styles, groups, types, and classes.

Style X - New
Style Y - Retreaded

Group 1 - Passenger car
Group 2 - Light Truck
Group 3 - Truck-bus

Type BA - Bias, tubeless
Type RA - Radial, tubeless
Type BB - Bias, tube-type (not applicable to Group 1)
Type RB - Radial, tube-type (not applicable to Group 1)

Class r/t - regular tread
Class a/s - all season tread
Class s/t - snow tread
Class a/t - all terrain tread (not applicable to Group 1)
Class o/o - on/off road tread (not applicable to Group 1)
Class d/a - drive axle tread (not applicable to Group 1)
Class t/o - trailer only (not applicable to Groups 1 and 2)

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1.2.2 SIZE AND LOAD DESIGNATION. Tires shall be of the sizes and load designations listed in "The Tire and Rim Association, Inc. Yearbook" or "The European Tyre and Rim Technical Organisation Yearbook".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to extent specified herein.

FEDERAL SPECIFICATIONS:

ZZ-I-550: Inner tube, pneumatic tire.
ZZ-T-441: Tire, Pneumatic: Retreaded and Repaired

FEDERAL QUALIFIED PRODUCTS LIST:

QPL-ZZ-T-381: Tires, pneumatic, vehicular (highway).

FEDERAL STANDARDS:

FED. STD. NO. 123: Marking for domestic shipment (civil agencies).

49 CFR, FED. MOTOR VEHICLE SAFETY STD. AND REGULATIONS, SUPPLIMENT 11, dated 1984, Section 575.104: Uniform Tire Quality Grading Standards, Appendix C.

49 CFR 571, FED. MOTOR VEHICLE STD. 109, 110, 119, and 120.

49 CFR 574 - Tire Identification and Recordkeeping Regulation.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; Seattle, WA.

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

MILITARY SPECIFICATIONS:

MIL-T-4: Tire, Pneumatic, and Inner Tube, Pneumatic Tire: Tire With Flap; Packaging and Packing Of.

MILITARY STANDARDS:

MIL-STD-105: Sampling procedures and tables for inspection by attributes.

MIL-STD-129: Marking for shipment and storage (non-aircraft).

MIL-STD-1224: Visual Inspection Guide for Pneumatic Tires (non-aircraft)

MIL-STD-45662: Calibration Systems Requirements.

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

DEPARTMENT OF TRANSPORTATION:

49 CFR 571 - Motor vehicle safety standard Nos. 109, 110, 117, 119, and 120.

49 CFR 574 - Tire identification and recordkeeping regulation.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 OTHER PUBLICATIONS. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE TIRE AND RIM ASSOCIATION, INC. YEARBOOK: (Copies may be obtained from the Tire and Rim Association, Inc., Crown Pointe, Suite 150, 175 Montrose W. Ave., Copely, OH 44321)

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THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION YEARBOOK: (Copies may be obtained from the European Tyre & Rim Technical Organization, The General Secretary, ETRTO, Avenue Brugmann 32, Boite 2, B-1060, Brussels, Belgium).

JAPAN AUTOMOBILE TIRE MANUFACTURERS' ASSOCIATION, INC.: (Copies may be obtained from the Japan Automobile Tire Manufacturers', INc., 9th Floor, Toranomon Bld., No. 1-12, 1-Chome Toranomon, Mina To-Ku, Tokyo, Japan.

DEUTSCHES INSTITUTE FUR NORMUNG: (Copies may be obtained from Deutsches Institute Fur Normung, Burggrafenstrasse 4-10, Postfach 1107, D-1000, Berlin 30.

BRITISH STANDARDS INSTITUTION: (Copies may be obtained from the British Standards Institution, 2 Park Street, London W1A 2BS, England.

3. REQUIREMENTS

3.1 QUALIFICATION. Tires furnished under this specification shall be tires which have been qualified and listed on, or approved in writing by the qualifying activity for listing on the applicable Federal Qualified Products List. Qualification and listing of tires on the Qualified Products List does not guarantee the acceptance of the tires in any future procurement nor constitute waiver of the requirements of the specification. The cost for performing qualification or requalification tests shall be wholly borne by the applicant. The costs shall be paid in advance and include direct and indirect costs. Tires, tubes, and flaps shall be supplied for testing at no cost to the Government.

Only the regular (r/t) or all season (a/s) tread pattern tires shall be submitted for qualification testing. Qualification of either regular tread (r/t) or all season tread (a/s) class tires of a particular brand shall constitute qualification of other classes, provided that the tires are of equivalent construction with the exception of tread pattern and compound.

For Style X tires, qualification of a particular model tire of which the manufacturers quality is clearly established will be extended to other tire models of the same brand being of equal or higher quality as certified by the manufacturer and provided they are equivalent or better construction and materials.

For Style Y tires, qualification of a particular retreaders process for which the retreaders quality is clearly established will be extended to other tire tread classes retreaded by the same process, of the same tire group and tire type having equal or better compound and treadwear as certified by the retreader.

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For both Style tires, qualification may be extended to a tube-type tire brand or model, provided it is of equivalent construction (design and materials), with the exception of the bead, to the tubeless model qualified. Using this same provision, the tube-type shall also qualify the tubeless model or brand.

For both styles of tires, qualification may be extended to other manufacturing facilities for a particular tire group provided the manufacturer certifies that the process and quality control procedures are equivalent at the additional facilities.

For Style X tires, prospective suppliers shall submit a list of their tire sizes, brands/models, Type, and Class (see 1.2.1) and fabric materials of the body plies and belts or breakers of each group the manufacturer intends to furnish under this specification. A Government representative shall select the tires for laboratory, (when specified), and road qualification tests.

For Style Y tires, prospective suppliers shall submit a list of their retreading processes, as defined in Section 3.4 of Federal Specification ZZ-T-441H, tire sizes, models, Type, and Class (see 1.2.1) of each group the manufacturer intends to furnish under this specification. A government representative shall select the tires for laboratory, (when specified), and road qualification tests.

The Government representative shall inform the respective manufacturers which tire sizes have been selected for the road tests, and furnish the consignee's address for the shipment of the sample tires to be tested. The manufacturers shall arrange to supply test tire sizes selected they propose to qualify for listing on the Federal Qualified Products List. The tire samples shall be selected by a Government representative from the manufacturer's production line, after passing their final finish inspection or when tires to be selected are stored in a warehouse, the tire sample shall be selected, at random, from a batch of not less than 100 tires for Style X and 25 for Style Y. All tires selected for qualification tests shall be permanently identified by a tamper-proof seal to be furnished and applied by the Government representative.

When the sample tires are selected from the manufacturer's production line, the Government representative shall select the tire samples needed from a normal production run. These tires shall not represent special run tires. Special run tires are those tires which are given extra attention other than that given to normal production tires by the manufacturer's quality control procedures. The Government representative shall require the manufacturer to certify in writing that the tires selected are not special run tires.

3.1.1 RETESTS. In the event of failure to pass the laboratory tests required in paragraph 4.3.1, the manufacturer shall be allowed a maximum of one retest. Retest tires shall be selected at the same time as the initial test tires. The acceptance and rejection criteria is shown in Table II.

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3.1.2 REQUALIFICATION. Once a tire manufacturers particular model/retreading process of tire has been qualified, it shall remain qualified and be listed on the QPL for a period of 5 years for Group 1, 6 years for Group 2, and 8 years for Group 3. At the end of the appropriate time period, the manufacturer must requalify the original tire model/retreading process which was submitted for qualification or any other tire model for Style X to which qualification was extended based on the qualification of the original tire model. For Syle Y, the retreader must re-certify additional retreading facilities as specified in 3.1, paragraph 6.

The government shall require a supplier listed on the Qualified Products List to show cause why his tires/retreading process should remain on the Qualified Products List if he has modified his product or processing sufficiently that the validity of the previous qualification is questionable, or when deemed that the quality of the product is not being maintained. When it is determined that the product delivered does not meet the specification or that the product delivered differs from that originally qualified, the government shall give that manufacturer thirty days notice of the intent to remove the product from the qualified products list and inform the manufacturer of the reason thereof. If the manufacturer does not make satisfactory response within thirty days, the product shall be removed from the qualified products list.

When a tire model or retreading process is determined to be hazardous in use, the government shall immediately notify the manufacturer and the tire model or retreading process shall be removed from the qualified products list. The tire model or retreading process shall not be reinstated until the manufacturer satisfies the government that the hazardous condition has been corrected.

3.1.2.1 REQUALIFICATION TREAD WEAR TESTS. The tread wear requirement for requalification shall be the maximum allowable percent skid depth loss established by the tread wear performance of the particular type tires to qualify to the current Federal Qualified Products List. Control tires shall be used to establish the tread wear qualification point. Control tires are selected by the Government representative utilizing the results from the previous qualification testing.

3.1.2.2 REQUALIFICATION LABORATORY TESTS. Requalification laboratory test shall be conducted in accordance with paragraph 4.3.1.

3.2 MATERIALS.

3.2.1 COMPOUND. The basic compound used in the tire construction shall be of natural rubber, synthetic rubber, or a combination thereof. Reclaimed rubber may be used in basic compounds. The manufacturer shall provide certification of the materials used in the construction of the basic compound to include any recovered materials.

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3.2.2 RETREADABILITY. Tire construction shall be of a type that allows for the retreading of the tire casing.

3.2.2.1 STYLE Y, CASING REQUIREMENTS: All tire casings furnished to the United States Government which have been retreaded shall be marked with the symbol "DOT" and meet specified requirements of Federal Motor Vehicle Standards 109 and 110, 117, or 119 and 120, as applicable.

3.2.3 REGULATORY REQUIREMENTS. In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

3.3 TIRE UNIFORMITY.

3.3.1 BALANCE LIMITS. Tires shall meet the applicable limits shown in Table I.

3.3.1.1 CORRECTIVE METHODS. Tires exceeding the balance limits in Table 1 may be used if corrected with balance pads, permanently affixed to the inside of the tire. Pads shall be so designed and placed as to prevent chafing of an inner tube when installed.

3.3.2 FORCE VARIATION. Group 1, Type RA tires shall have force variations as follows: The radial composite peak to peak force shall not be more than 45 pounds. The uncorrected radial first harmonic shall be not more than 35 pounds. The corrected (after grinding) radial first harmonic shall be not more than 25 pounds. The lateral composite peak force shall not be more then 28 pounds.

3.3.3 CONICITY LIMIT. Group 1, Type RA tires shall have a conicity limit of not more than 30 pounds.

3.3.4 RADIAL RUNOUT. Radial runout at the center rib of the tire tread of type RA and RB (class r/t only) tires shall not be more than .05 inch for Group 1, .08 inch for Group 2, and .09 for Group 3.

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TABLE I
TIRE BALANCE LIMITS
(IN.-OZ.)

GROUP	RIM DIAMETER	TIRE DESIGN SECTION WIDTH	MAXIMUM IN.OZ. HIGHWAY TIRES (Class r/t, a/s, t/o)	MAXIMUM IN.OZ. ON/OFF ROAD TIRES (class s/t, a/t, o/o, d/a)
1	All	8.00” or smaller	30	N/A
	ALL	8.01” or larger	35	N/A
2	13.0”	All	60	75
	14.0”	All	60	75
	14.5”	8.00” or smaller	75	105
		8.01” or larger	105	135
	15.0”	8.00” or smaller	60	75
		8.01” to 12.00”	75	105
		12.01” or larger	105	135
	16.0”	8.00” or smaller	60	75
		8.01” to 9.00”	75	105
		9.01” to 10.00”	105	135

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		10.01” or larger	135	175
	16.5”	8.00” or smaller	60	75
		8.01” to 10.00”	75	105
		10.01” to 12.00”	105	135
		12.01” or larger	135	175
	17.5”	8.00” or smaller	60	75
		8.01” to 10.00”	75	105
		10.01” or larger	105	135
3	15.0” to 18.0”	8.00” or smaller	75	105
		8.01” to 9.00”	105	135
		9.01” to 10.50”	135	175
		10.51” to 11.25”	175	215
		11.26” to 12.00”	215	265
		12.01” to 14.00”	265	315

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	14.01” or larger	315	365
19.5”	8.00” or smaller	105	135
	8.01” to 12.00”	135	175
	12.01” to 15.00”	175	215
	15.01” to 16.00”	215	265
	16.1” to 17.00”	265	315
	17.01” to 18.00”	315	365
	18.00” or larger	365	415
20.0”	9.00” or smaller	105	135
	9.01” to 10.50”	135	175
	10.51” to 11.25”	175	215
	11.26” to 12.0”	215	265
	12.01” to 14.00”	265	315
	14.01” or larger	315	365

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22.0”	10.50” or smaller	175	215
	10.51” to 12.00”	215	265
	12.01” to 14.0”	265	315
	14.01” to 16.00”	315	365
	16.01” or larger	365	415
22.5”	8.00” or smaller	105	135
	8.01” to 10.50”	135	175
	10.51” to 11.75”	175	215
	11.75” to 14.00”	215	265
	14.01” to 16.00”	265	315
	16.01” to 17.00”	315	365
	17.00” or larger	365	415
24.0”	11.00” or smaller	215	265
	11.01” or 13.00”	265	315

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	13.01” to 15.00”	315	365
	15.01” or larger	365	415
24.5”	10.50” or smaller	175	215
	10.51” to 12.00”	215	265
	12.01” to 14.00”	265	315
	14.01” to 16.00”	315	365
	16.01” or larger	365	415

3.4 PERFORMANCE.

3.4.1 Tires shall not show evidence of broken cords, tread chunking, groove cracking, separation of tread, plies or bead, when tested in accordance with 4.3.2.

3.4.2 OZONE RESISTANCE.

3.4.2.1 All tires, as part of production, shall contain sufficiant anti-oxidants and anti-ozonents to provide standard commercial resistance to weathering.

3.4.3 TEMPERATURE ABILITY. All tires supplied shall have an inherent capability of both storage and acceptable performance in ambient air temperature ranging from plus 120.0 degrees F (49.0 degrees C) to minus 40 degrees F (-40 degrees C).

3.5 AGE OF TIRES. Tires furnished under this specification shall not be more than eighteen (18) months old from the date of manufacture for Style X tires, or eighteen (18) months old from the date of retreading on the date the tires are delivered.

3.7 SPECIAL LABELING (see 6.2). When specified, each tire shall have a special label on the face of the tread. This label shall supplement the manufacturers’ commercial label, so when combined they show the tire size, Style, Group, Type, Class, and load designation or ply rating

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whether tube type or tubeless, tread type, ply material, (i.e., nylon, rayon, or polyester, etc.), the National Stock Number, Contract Number, Purchase Order Number, the month and year of manufacture, and the average weight. The label shall have a pressure sensitive adhesive backing which will not allow accidental loss and will not cause deterioration of the tread compound. All printing shall be clear and readable and shall boldly contrast with the label's background. The National Stock Number shall be in letters and numbers not less than 1/4 inch high. The special label and the manufacturers commercial label shall be placed on the tread face so that both may be read at the same time by one individual without having to rotate the tread.

3.8 WORKMANSHIP. Tires furnished under this specification shall show no evidence of poor workmanship. All plies, belts, bead, and breakers shall be smooth and evenly laid and shall be free of buckles, wavy cord, air pockets, and any other defects or imperfections which may impair serviceability. Casings for Style Y tires shall conform to the requirements of paragraph 3.3 of Federal Specification ZZ-T-441.

4. QUALITY ASSURANCE PROVISION

4.1 RESPONSIBILITY FOR INSPECTION. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for performing all the inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facility suitable for the performance of the inspections, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the tires/retreading process conform to prescribed requirements.

4.1.2 CLASSIFICATION OF INSPECTIONS.

a. Qualification inspections shall consist of those tests and examinations specified in 4.3.1.

b. Conformance inspections shall consist of those examinations specified in paragraphs 4.1.3, 4.1.4, and 4.1.5.

c. Preparation for delivery inspection shall be as specified in 4.4.

4.1.3 SAMPLING FOR ACCEPTANCE INSPECTION. Sampling for acceptance inspection shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.4 INSPECTION LOT. The inspection lot shall consist of tires, relative to a purchase order, of one group, one size, and of a particular brand, model, construction, and ply material from an identifiable production period, from one manufacturer.

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4.1.5 VISUAL EXAMINATION. The "sample unit" shall be one completely fabricated tire. Visual examination shall be in accordance with MIL-STD-1224. For Style N tires, the AQL for major defects shall be 4.0 percent defective and 6.5 percent defective for minor defects. The inspection level shall be S-4.

4.1.6 INSPECTION AND MEASURING DEVICES. Calibration of all inspection and measuring devices shall be in accordance with MIL-STD-45662.

4.3 QUALIFICATION TESTS.

4.3.1 QUALIFICATION LABORATORY TESTS. When specified, the qualification laboratory tests shall be performed by the manufacturer and witnessed by a representative of the Government at the prospective supplier's or other Government approved commercial laboratory. The laboratory qualification test shall consist of balance limits in accordance with 3.3.1 and Table I, force variations for only radial ply tires in accordance with 3.3.2 and Table II, and radial runout in accordance with 3.3.4 and Table II. The laboratory tests shall be performed on equipment currently being used by the tire manufacturer. In the event of a sample tires failure to pass the laboratory tests, the manufacturer shall be allowed a maximum of one retest.

TABLE II

Laboratory Tests			Number of Failures			
			Tests		Retest	
Total Quantity Tires for Tests	Characteristics	Total Test	Acc.	Rej.	Acc.	Rej.
3	Balance Limits	3				
	Force Variations	3				
	Conicity Limits	3				
	Radial Runout	3				
	Accumulative Tests	12	2*	3	1	2

*Two tires failing the same test characteristics is a reject.

4.3.2 TREAD WEAR AND CARCASS DURABILITY ROAD TEST. The treadwear and durability road tests shall be performed at a commercial test facility acceptable to the Government or at Government test facilities as specified. The test may be witnessed by a Government representative.

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4.3.2.1 VEHICLES. The vehicles used for testing a particular group of tires shall be of the same manufacture, model and model year, have identical drive trains (engine CID and horsepower, transmissions, and rear axle ratio) brake shoes/disc pads and drums/rotors, suspension components, and etc., and be of identical curb weight. All vehicles used for testing shall be rear wheel drive. For Group 3 testing, a tandem axle tractor with a tandem axle flat bed trailer shall be used as the test vehicle. Test tires shall not be used on the steerable wheels of the tractor or the inside position of dual wheels if the test vehicle is equipped with same. The load on each wheel shall not be less than 85 percent or more than 90 percent of the maximum single load of the test tire for Groups 1 and 2, and not less than 90 percent or more than 95 percent for Group 3, as specified in The tire and Rim Association Inc. Yearbook, issue in effect. After the vehicles are correctly loaded, they shall stand for 24 hours. If the suspension bottoms out during this period, the vehicle shall not be used to conduct the test. A constant weight shall be maintained in the drivers seat area that will equate the driver to that of the heaviest driver in the caravan. All axles of the vehicles shall be aligned in accordance with the vehicle manufacturers specifications prior to the beginning of the test. This alignment shall be checked at each tire rotation period. Any change in the vehicle or wheel alignment outside the manufacturers specifications shall be recorded and corrected. Vehicles shall be inspected before each test period for loose steering linkage, loose suspension components, loose wheel nuts, damaged rims, etc. Lateral or radial runout shall not exceed 0.034 inch for passenger car rims, 0.055 inch for light truck rims, and 0.065 inch for truck-bus rims. If a defect is noted, it shall be recorded and corrected. Tires of different sizes or construction shall not be mixed on any test vehicle. Tires of each Group shall be mounted on the test vehicles as singles, one tire on each wheel position. Tires of each group shall be tested on vehicles in caravan running at the same time, with vehicles a safe distance apart while still maintaining visual contact with all other vehicles in the caravan. Tires are to be tested on identical routes to assure that each tire is tested under the same climatic and road conditions. There are to be no changes to a route once it has been established and the test has begun.

4.3.2.2 MOUNTING. The tires shall be mounted with tubes (when required) meeting the requirements of ZZ-I-550 (see 2.1) and flaps (if required) on rims specified in The Tire and Rim Association Inc. Yearbook, issue in effect. The tire shall be inflated to the applicable pressure for the maximum load as specified in The TRA Yearbook with air filtered to remove all, dirt and water. Any pieces of rubber (vents and/or flash, etc.) remaining on the tire tread resulting from improper trimming of the tire during the final finish operation shall be removed and tire and wheel shall be dynamically balanced for Groups 1 and 2, and statically balanced for Group 3, and skid depth measured (see 4.3.2.5.1).

4.3.2.3 TEST PROCEDURE, GROUP 1.

4.3.2.3.1 PROCEDURE. The qualification test for groups 1 shall consist of 20,000 miles (+/- 20 miles) of operation for each tire. The first 800 miles shall be a break-in period. The test surface shall be free of ice and snow. Vehicles shall be operated over the same route at the same speed. In no event shall the test vehicle exceed the maximum prescribed, posted, or otherwise

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regulated speed limit for the public way upon which the tests are being conducted. All vehicle operators will take care to avoid all impaired or damaged shoulder areas of highway except in emergency cases. At least 85 percent of the test shall be conducted at not less than 50 miles per hour (MPH) nor more than 65 MPH for a particular tread wear test. One day test period shall consist of not less than 400 miles. Any test tire failing the road test as a result of a road hazard, prior to completion of 90 percent of the test mileage, may be replaced by the spare and the balance of the test miles run on that tire. One failure other than road hazard shall disqualify the tire.

4.3.2.3.2 ROTATION PROCEDURE. The tire rotation period for each particular group shall be at each 800 mile increment of the test (25 rotation periods). Each test tire shall be rotated to a different wheel position at each tire rotation period of the test. Each test tire shall be rotated to another vehicle after being mounted on two positions of a vehicle. Each test tire is to be rotated to a different wheel position at each prescribed rotation period. The rotation pattern is to be a rearward-cross pattern in which the tires on the forward axle of the test vehicle are moved to the rear axle of that vehicle on the opposite side. Tires on the rear axle are to be rotated to the forward axle of the next test vehicle in the convoy on the same side. Tires on the rear axle of the last test vehicle in the convoy are to be moved to the front most axle on the same side of the first test vehicle in the convoy. When only a single test vehicle is involved, the tire rotation shall continue on that vehicle for the duration of the test. The test vehicles shall remain in the same order in the caravan throughout the entire test for any particular group of tires.

4.3.2.4 TEST PROCEDURE, GROUP 2 AND 3.

4.3.2.4.1 PROCEDURE. The qualification test for Groups 2 and 3 shall consist of 20,000 miles (+/- 20 miles) of operation for each tire. The first 800 miles shall be break-in period. Ten percent (2,000 miles) of the test for Group 2 and five percent (1,000 miles) of the test for Group 3 shall be conducted over rough gravel secondary roads (pea gravel not acceptable), free of ice and snow, and be incorporated into each rotation period (800 miles) test run evenly; (80 +/- 0.5 miles for Group 2 and 40 +/- .5 miles for Group 3 of gravel road operation for every 800 mile period). At least 85 percent of the gravel road testing shall be conducted at 30 miles per hour (MPH) to the maximum extent possible. The remaining 18,000 miles for Group 2 and 19,000 miles for Group 3 (+/- 20 miles) of the test shall be conducted on paved highways or paved surfaces free of ice and snow. At least 85 percent of the paved surface test shall be conducted at no less than 50 miles per hour (MPH) and no more than 65 MPH. Vehicles shall be operated over the same route at the same speed, to the maximum extent possible. In no event shall test vehicles exceed the maximum prescribed, posted, or otherwise regulated speed limit for the public way upon which the test is being conducted. Rotation procedure shall be the same as described in paragraph 4.3.2.3.2.

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4.3.2.5 QUALIFICATION EVALUATION METHOD.

4.3.2.5.1 TREAD MEASUREMENT. A computation shall be made and recorded of the tread loss (percent skid depth loss) of the tires submitted for qualification testing. The tread loss shall be determined from skid depth measurements to the nearest 0.001 inch taken in eight (8) locations in each groove with all grooves measured at the same lateral point across the tire tread as located by a mark on the sidewall. This result shall then be averaged. Measurements shall be made at these same 8 locations throughout the duration of the test. Measurements shall be taken every 3,200 miles and shall begin following the 800 mile break in period. Measurements shall not be made on the treadwear indicators. A minimum of four (4) grooves shall be used (one in each shoulder and two in the crown) to evaluate the tread loss. If a test tire does not have at least three major grooves with one of these grooves within 1/2 inch of the centerline or at least four major grooves with two of these grooves located so that their outermost point is within ten percent of the S.70 of the centerline, the manufacturer shall mold or cut eight measuring voids in the area not meeting the above requirement. The eight measuring voids shall have a minimum depth of that of the deepest groove on the particular tire. The eight measuring voids shall be equally spaced around the circumference of the tire. Each measuring void may be placed on either side of the centerline of the tire. The molded or cut-in measuring voids shall be crescent shaped. Stud hole type voids are not acceptable. Minor cracking or cutting associated with these voids shall not result in a test failure.

NOTE: Measuring voids are only required for the tires submitted for testing and shall not be present on tires supplied under contracts.

NOTE: No test tire sample shall contain flash or extra rubber within the grooves that shall prohibit the measuring of the true tread depth by the measuring probe.

4.3.2.5.2 PROJECTED MILEAGE.The projected milage of the test tires shall be determined by:

1. Apply the method of least squares, as described in the Federal Motor Vehicle Safety Standards and Regulations, Supplement 11, dated 1984, Section 575.105: Uniform Tire Quality Grading Standards, Appendix C, using the average tread depth measurements described in 4.3.2.5.1 and the corresponding milages as data points, to determine the estimated regression line of Y on X.

$$Y = a + \frac{bx}{1000}$$

Y = average tread depth in mils

a = y intercept of regression line (reference tread depth) in mils, calculated using the method of least squares

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b = the slope of the regression line in mils of tread depth per 1,0000 miles, calculated using the method of least squares. This slope will be negative in value. The tires wear rate is defined as the absolute value of the slope of the regression line.

x = test miles

2. Determine the projected milage using the following formula:

$$\text{Projected Milage} = 1000 \frac{(a-62)}{b}$$

a = y intercept of regression line (reference tread depth)

4.3.3.5.3 QUALIFIED TIRES. All tires tested with projected milage equal to or higher than that of the projected milage of the established control tire shall be considered qualified for listing on QPL ZZ-T-381, issue in effect. All tires tested with projected milage lower than 95 percent of the established control tire shall be considered not qualified.

4.3.4 PURSUIT AND EMERGENCY HIGH SPEED TIRE QUALIFICATION METHOD, STYLE X ONLY.

Tires for this category shall not be road tested. All pursuit and emergency high speed tires shall be laboratory tested in accordance with 4.3.1. and Economic Commission for Europe (ECE) Regulation 30 test procedures or Society of Automotive Engineers J1561. Tires for this category shall be certified by the manufacturer as being T category (118mph), U category (124mph), H category (130mph), or V category (149mph) speed rated tires. Tires for this category shall be tested at a U.S. Government approved test facility properly equipped and staffed to perform testing under one or both of the test methods cited above. The tire manufacturer shall supply to the Government the test results, by tire model and size, and certification that the particular tire model and size successfully met the requirements of the subject tests.

Only those tires meeting the above requirements shall be qualified for listing on the qualified products list.

4.4 EXAMINATION OF PREPARATION FOR DELIVERY REQUIREMENTS. Packaging, Packing, and Marking Requirements shall be examined for conformance with Section 5. Sampling shall be in accordance with MIL-STD-105, inspection Level 4. The AQL shall be 4.0 percent defective. Any deviation from the specified preparation for delivery requirements shall be considered a defect.

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5. PREPARATION FOR DELIVERY.

5.1 PACKAGING. Packaging shall be Level A, B, or C of MIL-T-4J as specified (see 6.2)

5.2 PACKING. Packing shall be in accordance with MIL-T-4J.

5.3 MARKING. In addition to the marking requirements of the contract or order. The tires and shipping containers shall be marked in accordance with Federal Standard No. 123 for Civil Agencies, and MIL-STD-129 for Military Agencies.

6. NOTES.

6.1 INTENDED USE. Tires covered by this specification are intended primarily for use on passenger cars, pursuit and emergency passenger vehicles, station wagons, light, medium, and heavy trucks, buses, recreational, heavy load carrying and all types of vehicle trailers, and special purpose vehicles used by the Government.

6.2 ORDERING DATA. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Style, Group, Type, Class, size, and load designations (see 1.2.1 and 1.2.2)
(c) Speed Rating (if high speed pursuit type tire)
(d) Inspection and Testing Responsibility (4.1)
(e) Selection of Applicable Level of Packaging and Packing required (see 5.1 and 5.2).

6.5 DEFINITIONS.

6.5.1 ALIGNMENT. Adjustment of various parts of the vehicles suspension system to ensure proper handling stability and to minimize abnormal tire treadwear.

6.5.2 BIAS PLY TIRE. A bias ply tire is a pneumatic tire in which the ply cords that extend to the beads are laid at alternate angles substantially less than 90° to the centerline of the tread.

6.5.3 CLASS R/T: Regular tread highway tires, general purpose, all wheel position, rib pattern, for use on highways. This pattern is ideally suited for year-round use in southern U.S. and for spring, summer, and fall in northern U.S. This tread pattern also provides adequate traction in mild winter conditions.

6.5.4 CLASS A/S: All season tires are equipped with treads initially designed for use all year round, on all wheel positions of a vehicle, on improved highway surfaces (turnpike or city driving). This tread pattern provides adequate traction in mild snow conditions.

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6.5.5 CLASS S/T: Snow tires are equipped with block/lug type treads initially designed for service over improved snow covered highways.

6.5.6 CLASS A/T: The all terrain tread pattern is an aggressive lug pattern for use on vehicles where a substantial portion of the driving will be off-road on trails and cross country type terrain. It provides good off-road performance with respect to mud, sand and snow traction.

6.5.7 CLASS O/O: The on/off road tread pattern provides traction for vehicles used extensively on dirt and gravel roads. The pattern is designed so that it will not retain rocks and other debris in the grooves. The pattern is offered for both steering and drive axles. The pattern provides good traction in mud and snow conditions encountered on secondary roads.

6.5.8 CLASS D/A: Tread design for year-round use on medium/heavy trucks/buses used primarily on the highway. This tread pattern provides higher mileage and better driving traction than regular tread tires but may offer less traction than snow tires in snow conditions.

6.5.9 CLASS T/O: Tires and tread patterns designed for use on trailers only.

6.5.10 LOAD IDENTIFICATION. Load Range (A, B, C, etc.) identifies a given size tire with its load and inflation limits. Standard load and extra load are used to identify a given size P-metric tire with its load and inflation limits.

6.5.11 LOAD DESIGNATION. Load designation refers to the load range, ply ratings, and standard load or extra load.

6.5.12 OVERALL TIRE WIDTH. The overall tire width is the linear distance between the exteriors of the sidewalls of an inflated tire, including elevations due to labeling, decoration, protector bends, and tire growth.

6.5.l3 PLANT. Plants are the new tire or retreaded tire producing facilities of a company located in one Geographical location.

6.5.14 RADIAL PLY TIRE. A radial ply tire is a pneumatic tire in which the ply cords, which extend to the beads, are laid at substantially 90° to the centerline of the tread.

6.5.15 TEST LOAD. The test load for Groups 1 and 2 is 85 percent + 5 percent and for Group 3 90 percent +5 of the load listed in The Tire and Rim Association Inc. Yearbook and is to be used for testing purposes only. It is not to be used as the maximum load carrying capacity of the tire.

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6.6 TIRE TERMINOLOGY.

6.6.1 ASPECT RATIO. The ratio of the section height to the section width of the tire.

6.6.2 BALANCING. A process to correct for heavy or light areas of a tire to improve performance.

6.6.3 BEAD SEPARATION. A breakdown of bond between components in the bead area of the tire.

6.6.4 BELT SEPARATION. A tire failure in which the belts separate from the plies or tread.

6.6.5 BLISTER. A separation within the sidewall or liner stock or a separation between sidewall carcass or liner and carcass of the tire.

6.6.6 BREAK. A crack extending into or through the fabric. An impact break is usually in the shape of an “X”, “Y”, or star and can be seen from the inside of the tire. A flex or circumferential break runs parallel to the bead.

6.6.7 BUCKLE. Tire distortion caused by improper molding, evidenced by wrinkling on the inside of the casing.

6.6.8 CHECKING. Minute cracking in surface of rubber caused by aging or oxidation.

6.6.9 CHUNKING. The breaking away of pieces of the tread.

6.6.10 CIRCUMFERENTIAL BREAKS. A tire injury running parallel to the bead.

6.5.11 CIRCUMFERENTIAL TREAD CRACKS. Cracks in the grooves of the tread running parallel to the beads.

6.6.12 TREAD. The tread is the portion of a tire that comes in contact with the road.

6.6.13 CORD. The cords are the strands forming the plies in the tire.

6.6.14 BEAD. The bead is the section of the tire made of steel wires, wrapped or reinforced by ply cords, that is shaped to fit the rim.

6.6.15 SIDEWALL. The sidewall is the portion of the tire between the tread and the bead.

6.6.16 GROOVE. Groove is the space between two adjacent tread ribs.

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6.6.17 CRACKING. Cracking is any parting within the tread, sidewall, or inner liner of the tire extending to the cord material.

6.6.18 OPEN SPLICE. Open splice is any parting at any junction of tread, sidewall, or innerliner that extends to the cord material.

6.6.19 CORD SEPARATION. A cord separation is the parting of rubber compound from the cord.

6.6.20 PLY SEPARATION. Ply separation is a parting of rubber compound between adjacent plies.

6.6.21 TREAD SEPARATION. Tread separation is the pulling away of the tread from the tire carcass.

6.6.22 SIDEWALL SEPARATION. Sidewall separation is the parting of the rubber compound from the cord material in the sidewall.

6.6.23 SECTION WIDTH. The section width is the linear distance between the exterior of the sidewalls of an inflated tire excluding elevations due to labeling, decoration, or protective bands.

MILITARY CUSTODIAN:
Army - AT

USER INTEREST:
Army - CE

REVIEW ACTIVITIES:
Navy - YD
Army - AT, WC

PREPARING ACTIVITY:
GSA - FSS

CIVIL AGENCY COORDINATING ACTIVITIES:
DOT - ACO, MVP
USDA - AFS

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